HOLY Marketing Analyst Case — Full Solution

Rodrigo Uehara Guskuma · July 2026

Code: all computations behind this document live in `analysis.ipynb` (rendered on GitHub with outputs) — the `(code §…)` tags throughout point to its numbered sections. Full repository: https://github.com/ugrodrigo/holy-case-rodrigo-uehara-guskuma

Contents

Part 1 — Q1: 5th BURRsday Syrup Campaign — Launch Analysis

*Briefing for the Head of Marketing · Data: Apr 1 – May 31, 2026 (DE/FR/UK Shopify) · All monetary figures are gross revenue and should be read as **relative** comparisons, per the dataset disclaimer.*

> **Data prep:** duplicate orders deduped, fully-cancelled orders excluded, orphan line items dropped, delivery delay computed as `delivered_at - order_date`. Details and open data questions in Appendix A — Data Quality Review.
>
> **Code:** every number in this document is produced by **analysis.ipynb** — each section carries a `(code §…)` tag pointing to the notebook section that computes it (cleaning rules in §2).

TL;DR — the five things to know

1. **The launch worked as a demand event.** Campaign revenue ran at **+124% per day vs. the pre-period** (+60% excluding launch day). Launch day alone did ~€3.4M — 6× a normal day — and half of all campaign bottle volume.
2. **It was an existing-customer launch, not an acquisition launch.** Only **22.9%** of syrup-bottle orders came from first-time customers vs. 47.9% for non-syrup orders in the same window. The syrup bottle monetized the base; it did not (yet) bring new people in.
3. **DE carried it**: 74% of bottle units, 56.8% of DE campaign orders contained a bottle (FR 30.3%, UK 22.3%). FR underperformed its size (+55% revenue uplift vs. DE +191%).
4. **The production gap was expensive.** Bottle orders placed in weeks 18–19 waited a **median 38–40 days** for delivery. Reorder rate drops with delay: **14.5% (delivered ≤4 days) → 5.4% (>14 days)** raw;

after correcting for calendar truncation the causal penalty is **~25% lower reorder likelihood, ≈1,000 lost reorders** (§5.6). It was also a **single-warehouse (POZ1) problem** — NOT1 shipped bottles on time throughout (§5.4).

5. **Early retention is real but shallow:** 10.1% of identifiable campaign bottle buyers re-bought syrup within ~4 weeks (median 23 days to reorder), mostly via the May 12 3er pod bundles — evidence the refill model works. No cannibalisation of the core Energy business is visible.

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1. Setting expectations

KPIs I would track for a launch like this

KPI	Why it matters
Adoption / penetration (% of orders containing the hero SKU; units)	Direct measure of whether the launch landed
Incremental revenue uplift vs. pre-campaign baseline (total and per market)	Separates "campaign grew the pie" from "campaign relabeled existing demand"
New vs. existing customer mix of hero-product buyers	Tells you whether the product acquires or monetizes
AOV & attach rate (pods, consumables, cross-category)	The syrup system is razor-and-blades: the bottle only pays off if consumables attach
Repeat / reorder rate of hero buyers (30/60/90-day cohorts)	The single most important long-term KPI for a consumable ecosystem
Fulfilment SLA (days to delivery, % delayed)	Launch experience drives second purchase; we can measure it directly
Category mix shift / cannibalisation	Guards the core Energy business
CAC / media efficiency *(not measurable here — no spend data)*	Needed to call ROI, flagged as an open ask

What "good" looks like

- **End of campaign:** high penetration among existing customers (the launch audience), strong pods attach (validates the system), demand sustained after the launch-day spike rather than a one-day novelty, and no meltdown in fulfilment.
- **+30 days:** the KPI shifts from adoption to **repeat**. Good = a double-digit share of bottle buyers back for refills within 30 days, refill volume replacing launch hype as the demand driver, and no lasting dent in Energy sales. A first-ever product has no benchmark, so the honest framing is: *set the benchmark now, manage the curve* (which is what the May 12 3er bundle did).

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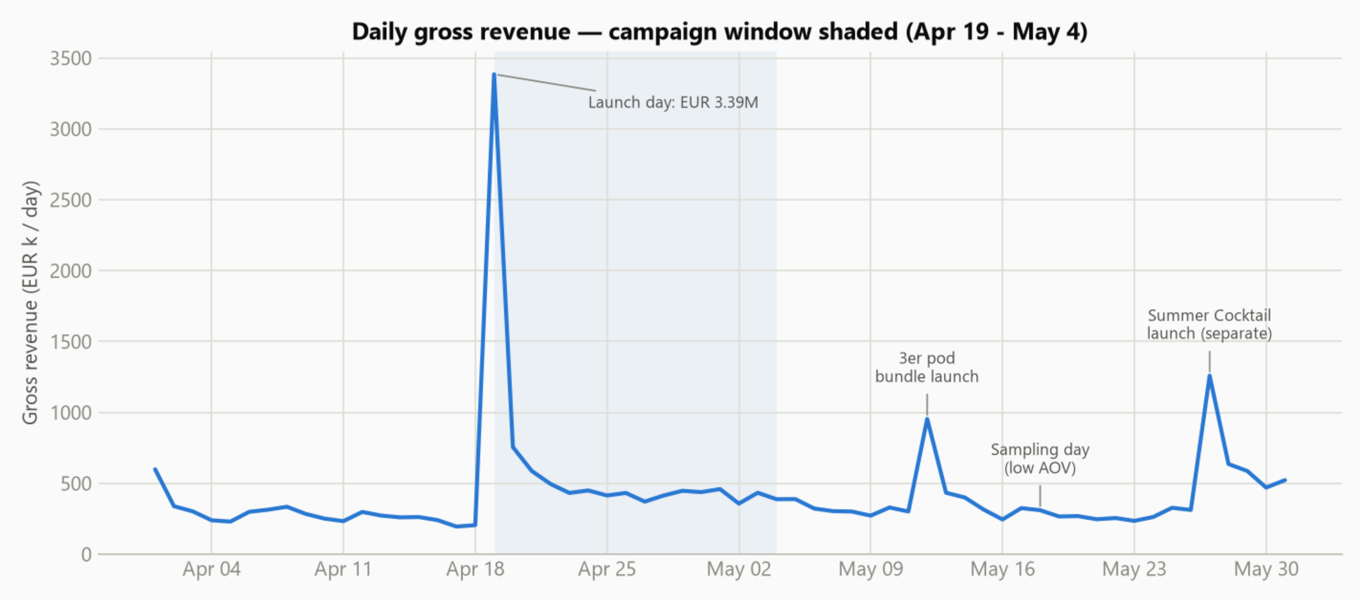
2. Campaign analysis

2.1 Baseline & uplift *(code §3.1)*



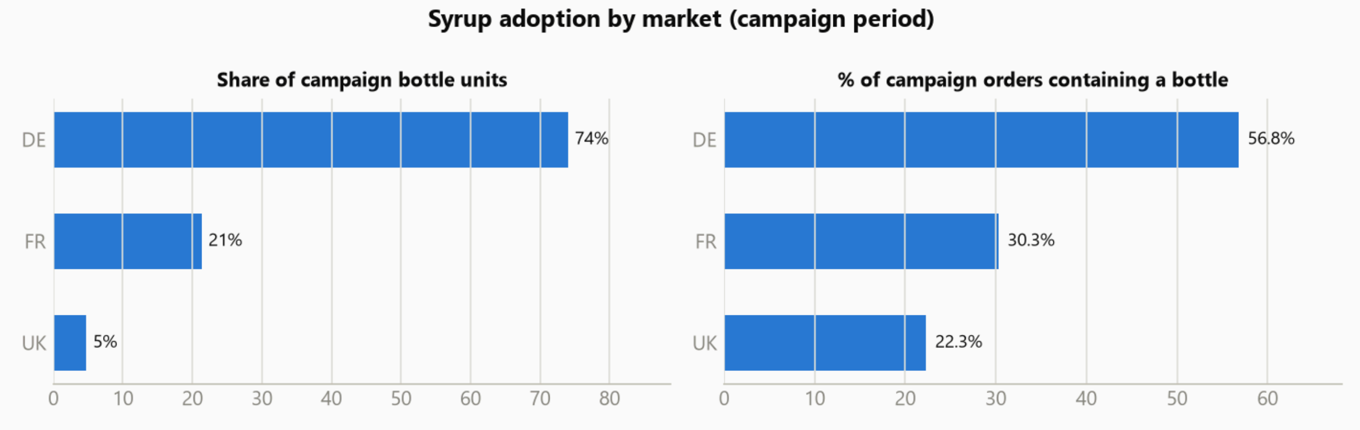
Period	Days	Orders/day	Gross €/day	AOV	First-order share
Pre (Apr 1–18)	18	4,046	€287k	€70.9	46.9%
Campaign (Apr 19–May 4)	16	**7,798 (+93%)**	**€641k (+124%)**	**€82.3 (+16%)**	36.8%
Post (May 5–31)	27	5,383 (+33%)	€402k (+40%)	€74.7	40.8%

- Excluding launch day, the campaign still ran **+49% orders / +60% revenue per day** — the uplift was not just one spike.
- The post-period stays ~40% above pre, but is **contaminated upward** by the May 12 3er launch, a May 18 sampling giveaway and the May 27 Summer Cocktail launch — read it as "elevated, partly for other reasons", not as pure campaign halo.
- **Interpretation (hypothesis):** launch-day concentration (28% of campaign orders in one day) points to a well-primed audience — likely CRM/community activation. Without media spend data we can measure *uplift*, not *ROI*.

2.2 Existing vs. new customers *(code §3.2)*

- Campaign first-order share **fell** to 36.8% (pre: 46.9%) — the extra demand was disproportionately **existing customers**.
- Bottle orders: **22.9% new** vs. 47.9% for non-bottle orders in the campaign window.
- **What the data shows:** the syrup launch activated the base. **My interpretation:** that's the right sequencing for a new format (fans forgive teething problems), but H2 needs an acquisition angle for syrup — currently it doesn't pull new customers.

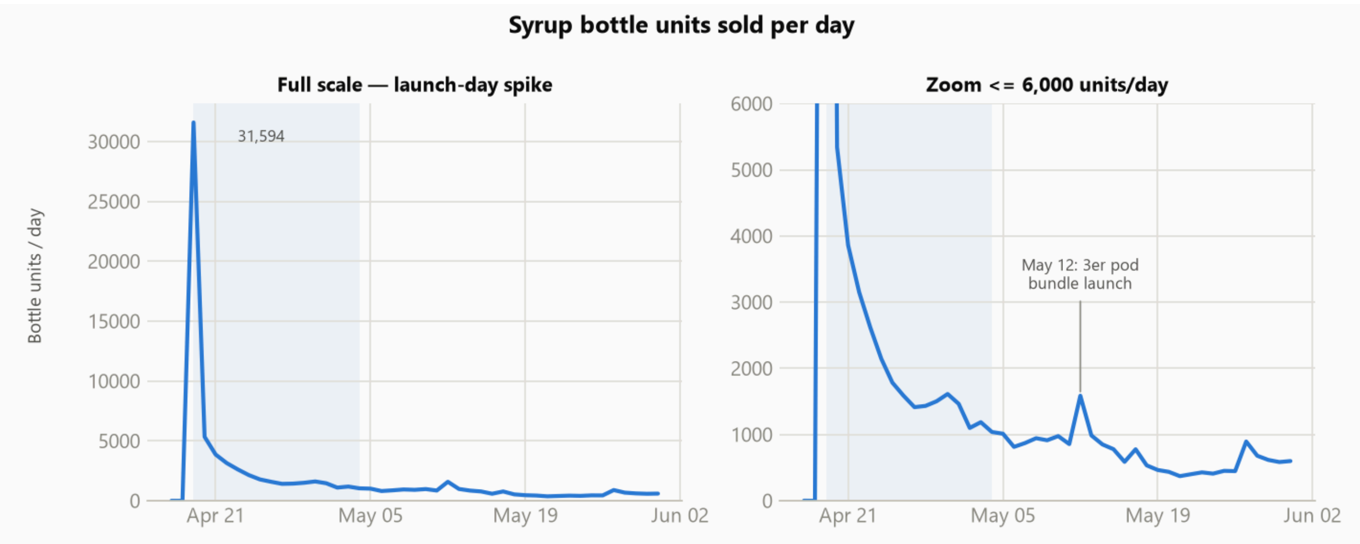
2.3 Market breakdown *(code §3.3)*



Market	Revenue uplift (campaign vs pre, €/day)	Bottle units share	Bottle penetration of campaign orders
DE	**+191%**	74%	56.8%
UK	+109%	5%	22.3%
FR	**+55%**	21%	30.3%

FR is the underperformer relative to its base size (FR is ~40% of pre-period revenue but took only 21% of bottle units). Hypotheses to check with the team: weaker localisation/creator support, price sensitivity, or the campaign being DE-centric by design.

2.4 Daily trend *(code §3.4)*



Launch day sold **31,594 bottles — 50% of total campaign bottle volume** — then decayed to a steady ~1,000–1,600/day for the rest of the campaign, settling at ~400–900/day post-campaign with a clear bump on May 12 (3er launch). The demand curve is a classic hype-then-baseline launch; the post-campaign bottle baseline (~500/day) is the number to plan production against.

2.5 Category mix *(code §3.5)*

Share of gross revenue by category:

Category	Pre	Campaign	Post
Syrup Bottle + Syrup Bundle (pods)	0%	**35.0%**	13.8%
Energy (bundle + sachet + box)	29.5%*	21.5%	28.6%*
Mixed Sachetbox	27.3%	14.5%	17.2%
Hydration Bundle	15.1%	10.3%	14.2%
Iced Tea Bundle	12.2%	8.7%	10.8%

**Energy rows shown as the share of the three energy categories combined; full table in the analysis script output.*

Syrup instantly became the #1 category during the campaign, and **retains ~14% of revenue after it** — a real second leg, not a stunt. Energy's *share* dipped during the campaign but its **absolute €/day grew** (€84.6k pre → €137.8k campaign → €115.2k post), so the dip is denominator effect, not decline (see cannibalisation, §4.2).

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3. Products deep dive

3.1 Adoption & buying behaviour *(code §3.6, §3.7)*

- **44.5% of all campaign orders (55,487 of 124,767) contained a syrup bottle**; 62,836 bottles sold in 16 days.
- Flavour split: **Darko 49% · Syru 34% · Raptor 17%** — consistent enough that no flavour flopped, skewed enough to plan production toward Darko.
- **Every bottle was sold as part of a bundle** (100% `is_bundle`), with a **99.8% pods attach rate** and 81% sticker attach: the hero offer was clearly *bottle + 10er pods (+ sticker)*, and standalone bottle sales were effectively not a thing. Only 93 pod packs were sold without a bottle during the campaign.
- 88% of bottle orders took exactly one bottle (mean 1.13) — one bottle per household; multi-bottle gifting didn't materialize.
- **Bottle orders spend more:** AOV €91.7 vs €74.7 for campaign non-bottle orders (+23%).

3.2 Cross-purchase *(code §3.8)*

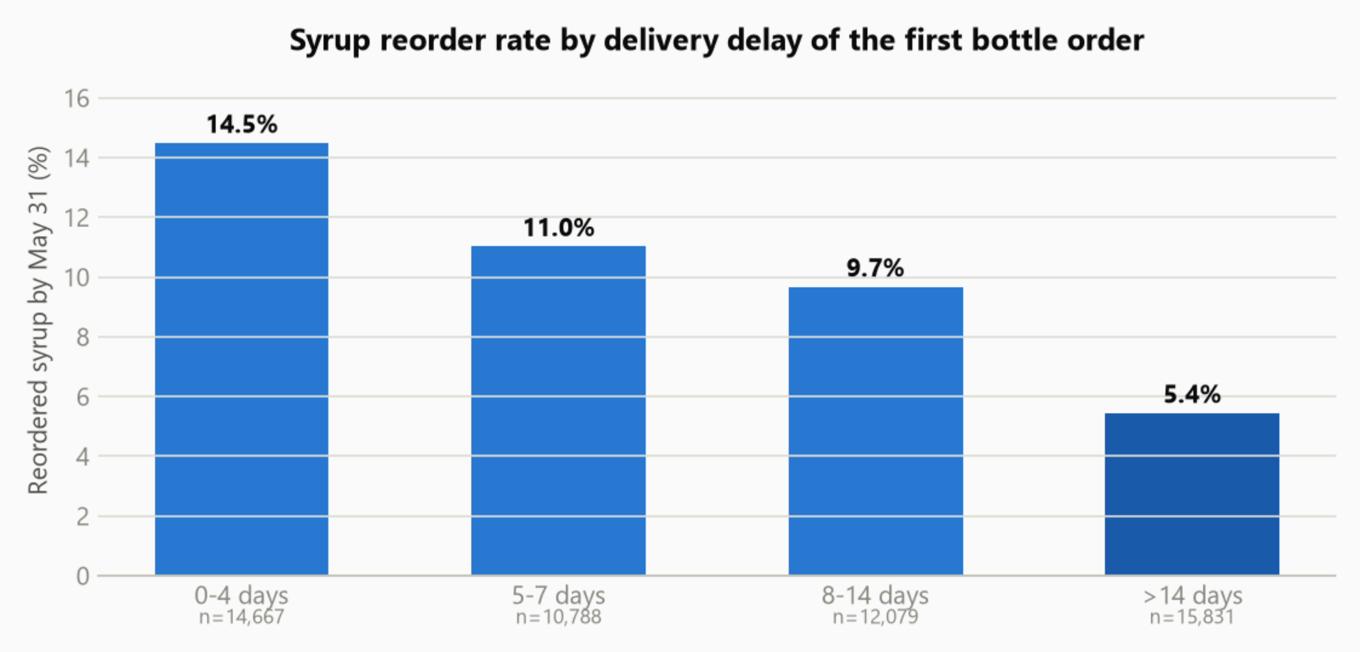
Among bottle orders (besides the bundled pods/sticker): **30% added a Shaker, 24% Merch, 16% an Energy tub**, 9% Iced Tea, 9% Hydration. Only 0.1% bought the bottle bundle alone. Syrup buyers are engaged multi-category customers — supports the "monetize the fans" read.

3.3 Delivery delay impact — the costliest lesson of the launch *(code §3.9)*

Bottle orders were dramatically slower than the rest of the shop:

	Median delay	% > 7 days	% > 14 days
Campaign bottle orders	**8 days**	**52.1%**	**29.8%**
Campaign non-bottle orders	4 days	14.7%	0.7%

By order week, bottle orders placed in **weeks 18–19 (Apr 27 – May 10) waited a median 38–40 days** — the production gap hit customers who ordered late in the campaign; service recovered to ~5 days by week 20.



Reorder rate falls monotonically with the delay on the *first* bottle order: **14.5% → 11.0% → 9.7% → 5.4%**. Holding the ≤4-day rate as the counterfactual, the naive estimate is **~2,400 lost reorder customers (~4.5pp of cohort retention)**. *Correction (see §5.6):* part of this gradient is calendar truncation — late-delivered customers had fewer days left to reorder before the data ends. A fixed 21-day-after-delivery window puts the **causal penalty at ~25% lower reorder likelihood, ≈1,000 lost reorders**. Still material, about half the naive read.

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4. After the campaign

4.1 Retention & reorders *(code §3.10, §3.12)*

Of **53,738 identifiable campaign bottle buyers**:

- **10.1% (5,417) re-purchased syrup by May 31** (window: 4–6 weeks); median time to reorder **23 days** — consistent with a ~3-week consumption cycle for the bundled 10er pods.
- What they reordered: **3er pod bundles 3,989 · bottles 2,014 · 10er pods 1,924** (overlapping). The May 12 3er launch was well-timed against the consumption cycle and immediately became the main reorder vehicle (3,374 orders on day one).
- 3er bundle buyers since May 12: 10,236 orders, only **9.3% first-time customers**, and **42.4% verifiably owned a bottle already** — it is functioning as a retention product, as intended. (The other ~58% may have bought bottles before Apr 1, via another channel, or are gift buyers — worth checking, and one of my interviewer questions.)
- Reorder flavour ranking (Peach, Cola Ice Pop, Green Apple on top) is the first read on which refill flavours to scale.

4.2 Cannibalisation risk *(code §3.11)*

No evidence of cannibalisation so far — if anything the opposite:

- Energy absolute revenue per day *rose* pre → post (+36%).
- Cohort comparison: existing customers who bought a bottle grew their energy spend/day **+74%** pre→post, vs **+52%** for existing campaign customers who didn't buy a bottle.

Interpretation with caveats: syrup looks complementary (different consumption occasion) rather than substitutive. But the post window is short, seasonal, and boosted by other launches; the honest verdict is "no early warning signs", to be re-checked on a 90-day horizon — the real risk is *share-of-stomach* over months, not weeks.

4.3 What worked / what to fix / open questions

Worked: launch-day activation of the base (28% of campaign volume in one day); bundle architecture (100% pods attach = every bottle owner starts the refill loop); the May 12 3er bundle timed to the refill cycle; a durable post-campaign syrup baseline (~14% of revenue).

Needs improvement:

1. **Fulfilment readiness** — the production gap measurably burned early retention (~1,000 reorders on the corrected estimate, §5.6) and it was POZ1-specific (§5.4); late-campaign buyers had a first experience of waiting 5+ weeks.
2. **FR performance** — 55% uplift vs DE's 191%; diagnose localisation/creator mix before H2.
3. **New-customer angle** — syrup didn't acquire; consider a syrup-led acquisition offer now that the base is saturated with bottles.

Couldn't answer with this data (and what I'd need):

- **Campaign ROI** → media spend & channel attribution for the window.
- **True incrementality of launch-day demand** (pull-forward vs. net-new) → longer pre-period / prior-year baseline.
- **Retention beyond 4–6 weeks & pods consumption rate** → June+ orders; subscription data if any.
- **Margin** impact of the bundle discount → COGS/discount data (net revenue in the extract is unreliable — see data review).
- Whether **delivery-delay victims** should get a win-back voucher (I'd A/B it — the 15k customers who waited >14 days are a named, reachable audience).

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5. Further analysis — beyond the brief's guiding questions

Seven analyses the case didn't ask for but that sharpen — and in one case honestly **revise** — the findings above.

5.1 Refunds & cancellations *(code §6.1)*

Refund incidence is tiny (~0.4% of campaign orders) and does **not** rise with delivery delay; cancellations are 2.7× higher for bottle orders (0.9% vs 0.3%) but still small. **The production gap's cost was retention, not immediate revenue give-back** — no product-quality red flag in refunds for a first-ever product.

5.2 The May 18 sampling giveaway barely converted *(code §6.2)*

The 5.4k new customers who received free samples on May 18 converted to a **paid order within 13 days at just 0.37%**, vs **4.8%** repeat-paid for organically acquired new customers in a matched window — ~13× worse. Short-term, the giveaway generated traffic, not customers. *(Caveat: 13 days is short; the leads may still monetise via CRM.)*

5.3 Campaign-acquired new customers are weaker, not stronger *(code §6.3)*

Fixed 21-day repeat window: campaign new customers **with** a bottle repeat at **4.5%**, vs 8.5% for campaign new customers without one and 9.3% for pre-period new customers. Even accounting for the ~23-day syrup refill cycle and the delivery delays, this further weakens syrup as an acquisition tool: it pulled few new customers (§2.2) and the ones it pulled repeat less — consistent with hype/gift buyers.

5.4 The production gap was a single-warehouse problem *(code §6.4)*

Bottle orders fulfilled from **POZ1 took a median 8 days (mean 13)**; **NOT1 shipped bottles in ~4 days throughout** — same as its normal service. The ops recommendation sharpens from "fix production planning" to a POZ1-specific stock-allocation fix (and consider cross-shipping from NOT1 during launches).

5.5 The launch re-activated dormant customers — and they stayed active *(code §6.5)*

Existing customers who bought a bottle nearly **doubled their total spend/day post-campaign (+88%)** while the matched control group *declined* (−5%). Notably, the bottle cohort was **less active pre-campaign** despite being larger — the launch pulled dormant fans back, and they kept ordering. *(Interpretation caveat: part of the lift is syrup refills themselves plus regression to the mean; a matched-activity control would isolate the pure habit effect.)*

5.6 Honest revision: the true delay penalty is smaller than the raw gradient *(code §6.6)*

The 14.5% → 5.4% gradient in §3.3 partly reflects **calendar truncation** (late-delivered customers had fewer days left to reorder before the data ends). Giving every customer the same **21-day clock starting at delivery** (deliveries ≤ May 10 only), the rates become **9.8% (0–4d) vs ~7.1–8.0% (delayed)** — a **~25% relative penalty**, or roughly **~1,000 lost reorders rather than the naive ~2,400**. The delay effect is real and material, but about half the naive estimate. TL;DR #4 and §3.3 should be read with this correction.

5.7 A refill-demand planning number *(code §6.7)*

From well-served owners' steady-state behaviour: **per 1,000 bottle owners, expect ~140 refill orders (~500 pod packs) per month**. First-month estimate from the launch cohort — refine with June+ data — but it's the number production planning lacked when the gap happened.

Time spent: ~4.5 hours end-to-end (data audit, analysis, charts, write-up, further analyses), with the analysis fully scripted in Python/pandas for reproducibility.

Part 2 — Q2: H2 2026 Sales Budget — Allocation, Testing Roadmap & Strategic Reflection

Inputs: MMM v5.3.1 memo (Section A) + DE attribution data Mar–May 2026 (Section B). Constraints: €35M total · €4M brand (fixed) · €800K TV locked in Q3 · €31M flexible performance budget.

> **Code:** the memo PDF extraction, its tables transcribed as data, and every derived number in this document are in **analysis.ipynb §5** — PDF extraction §5.1, memo tables §5.2, derived analysis & MMM-vs-attribution conflicts §5.3, allocation checksums §5.4.

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0. The elephant in the room (read this first) *(code §5.3)*

The MMM team's own optimised plan spends **€16.1M** in H2 — the task asks me to allocate **€35M**, i.e. **2.2× the model-optimal level** and ~1.8× the H1 run-rate (~€19.3M/half). The MMM says META's marginal ROI is already ~0.99 at *current* spend and TikTok's drops to 0.63 at just +50% spend. **At €31M of performance spend, the marginal euros are, per the model, value-destructive.**

I therefore treat the €35M as a hard constraint (growth/share-taking mandate, investor commitment, etc.) and build the plan that **minimises marginal-ROI damage, weights spend toward Q4** (Black Week/Christmas, when response curves support more spend), and **converts part of the budget into measurement** so the next planning cycle isn't flying blind. But my first recommendation as an analyst is explicit: *if the €35M is negotiable, right-size media to ~€20–24M and put the remainder into retention/CRM, promo depth, or reserves.*

Task 1 — H2 Budget Allocation

1.1 Allocation logic

1. **Rank by marginal ROI (mROI), not average iROAS** — iROAS rewards history; mROI prices the next euro. Ranking: Google 4.46 > Affiliate 1.78 > Paid Social 1.50 (TikTok 1.35 / META 0.99) > Brand 0.44 ≈ Influencer 0.40.
2. **Scale the small high-mROI channels first** (Google, Affiliate) — but capped, because the memo itself flags their numbers as attribution-assisted; scaling is gated by incrementality tests.
3. **Paid Social takes the largest share**, tilted to TikTok early and META for depth — accepting that both saturate below mROI 1 at this budget.
4. **Influencer is restructured, not just cut.** The MMM says the *mix* is the problem (YouTube Inf. 0.30 vs Podcast 0.71 / TikTok Inf. 0.76). Because €31M must be deployed, Influencer keeps a substantial envelope, shifted toward the high-mROI platforms as far as contract floors allow.
5. **Hold a testing & flex reserve** — with model holdout R^2 of 0.29, buying information is worth more than the last saturated media euro. The reserve deploys in Q4 into whatever the Q3 tests validate.
6. **Phase Q3 < Q4** (≈ 40/60): H2 contains Black Week + Christmas; saturation ceilings are higher when demand is elevated.

1.2 Channel-by-channel plan (H2 totals) *(code §5.4)*

Channel	H1 actual (≈26wk)	H2 plan	vs H1	Logic
Paid Social	€5.9M	**€17.0M**	+186%	Largest share; least-bad saturation profile. META €10.0M (workhorse, mROI ~1 → run to breakeven), TikTok €5.5M (momentum, front-scale but gated by the Aug stress test), YouTube paid €1.5M (scaling test of the "37× iROAS" signal — treated as unproven, capped, staged)
Influencer	€7.0M	**€9.6M**	+37%	Forced above optimal by the budget constraint. Mix shift: YouTube Inf. share cut from ~50% toward Podcast / TikTok Inf. / Instagram (mROI 0.5–0.8);

Channel	H1 actual (≈26wk)	H2 plan	vs H1	Logic
				renegotiate or exit floor-driven contracts; whitelisting/paid-amplification bridges Influencer → Paid Social
Google Search	€0.15M	**€1.0M**	~6.7×	Highest mROI, tiny base. Phase: 2× run-rate in Q3 (the memo's controlled test), scale to plan in Q4 only if the brand-keyword holdout confirms incrementality
Affiliate	€0.46M	**€0.9M**	~2×	mROI 1.78; scale moderately; audit coupon-poaching first
Testing & flex reserve	—	**€2.5M**	—	Funds incrementality tests + Q4 reallocation into validated winners (likely destinations: TikTok scale-up, Google, or promo support in Black Week)
Performance subtotal	€13.5M	**€31.0M**		
Brand (fixed)	€5.8M	**€4.0M**	−31%	See split below
TOTAL	~€19.3M	**€35.0M**	+81%	

1.3 Brand envelope (€4.0M) *(code §5.4)*

Allocated by the L2 iROAS ranking, with the memo's caveat that the 8-week window *understates* brand (so I don't zero-out anything structural):

Sub-channel	iROAS (8wk)	H2 plan	Logic
TV — Q3 (locked)	1.12	**€0.80M**	Contractual
Events / Sponsoring	**1.74**	**€1.50M**	Best measured brand ROI; scale
TV — Q4 top-up	1.12	€0.60M	Above water even on the conservative window; supports Q4 trading
OOH	0.97	€0.40M	~Breakeven on a lower-bound measure → hold, don't grow
Streaming	0.92	€0.25M	Same logic, smaller
Influencer Awareness	0.69	€0.30M	Cut hard; overlaps with performance influencer
Performance Awareness	0.43	€0.15M	Near-exit; keep a measurement stub
Total		**€4.00M**	

1.4 Quarterly phasing *(code §5.4)*

	Q3	Q4	Rationale
Performance	€12.0M	€19.0M	Black Week + Christmas raise saturation ceilings; Q3 runs the tests that de-risk Q4
Brand	€2.2M	€1.8M	TV lock + events season in Q3; Q4 brand supports peak trading
Total	**€14.2M**	**€20.8M**	

Task 2 — Testing Roadmap

Principle: every €1M+ allocation decision that rests on a disputed number gets an incrementality test *before* the Q4 reallocation gate (end of September). The tests target exactly the three places where MMM and attribution disagree most.

#	Test (priority order)	Hypothesis	Method	Timing	Decision trigger
1	**Influencer incrementality** — the biggest money on the most disputed number (MMM mROI 0.40 vs attribution CLV6M/CAC ≈ 100%)	≥50% of attributed influencer NCs would have converted anyway	Creator-level holdout: pause a matched ~30% panel of DE creators for 6 weeks; measure regional/audience NC delta vs control panel (geo-split is impractical for national creators)	**Jul–mid Aug**	Incremental CAC > €85 (≈1.5× attributed) → cut Influencer to floor in Q4, reserve moves to PS/Google. Incremental CAC ≤ CLV6M → restore mix-shifted budget
2	**Google Search brand-keyword holdout** — cheap, fast, independent	≥60% of brand-search conversions are non-incremental	Pause brand keywords in a matched set of DE regions (geo split) for 4 weeks	**Jul** (parallel to #1 — different mechanism, minimal interference)	True incrementality <40% → cap Google at 2× run-rate, iROAS 4.09 declared attribution artifact; ≥40% → scale to €1M+ plan
3	**META conversion-lift study**	mROI ≈ 1.0 at current spend (i.e. META is at breakeven, not under water)	Platform conversion-lift (ghost ads) on DE, 4–6 weeks	**Aug**	Lift-based iROAS <1 → freeze META at H1 level, shift to TikTok/reserve; ≥1 → hold plan
4	**TikTok scaling stress test**	mROI stays ≥1.0 up to +50% spend	Geo-lift: +50% spend in test regions vs control, 6 weeks	**mid Aug–Sep** (after #3 so PS geos aren't	mROI at scale ≥1 → deploy reserve into TikTok for Q4;

#	Test (priority order)	Hypothesis	Method	Timing	Decision trigger
		(MMM says it falls to 0.63)		double-treated)	<0.8 → hold TikTok at H1+20%
5	**Affiliate coupon-leak audit + commission holiday**	A meaningful share of affiliate revenue is last-click coupon poaching	Analytics audit + 2-week commission pause on coupon publishers in a segment	**Sep**	Sales unchanged during pause → restructure commissions before scaling
6	**YouTube paid staged scale-up**	The 37× iROAS is a small-base artifact but the channel clears mROI 1 at moderate spend	Budget doubling in capped steps with geo control (built into the €1.5M allocation)	**Jul–Oct, continuous**	Each step must hold lift-implied mROI ≥1.5 to unlock the next

Sequencing & interference rules:

- #1 (creator holdout) and #2 (search geo) run in parallel — different mechanisms and different randomisation units; monitor search-volume contamination in #2 regions from #1's social buzz reduction (direction would bias *against* finding search incrementality, i.e. conservative).
- #3 and #4 both manipulate Paid Social in DE geos → run **sequentially**, not simultaneously.
- **Test freeze from Nov 1** (Black Week + Christmas): no holdouts during peak; peak-season response data feeds the next MMM refresh instead.
- **Decision gate: end of September** — Q4 budgets (incl. the €2.5M reserve) reallocate based on tests #1–#4; results also become priors/calibration constraints for MMM v6.

Task 3 — Strategic Reflection

3.1 Assumptions I had to make (with confidence levels)

Assumption	Confidence
The €35M must actually be spent in media (no option to bank it)	Low — this is the assumption I'd challenge first
MMM response *curves* (saturation shapes) extrapolate to ~2× current spend	**Low** — the model was fit on €48M/125wk; €35M/26wk is far outside the training range
Channel *ranking* by mROI is directionally right even if point estimates aren't	Medium-high — this is exactly what the memo claims survives holdout R^2 0.29

Assumption	Confidence
DE attribution data generalises to the full budget scope (memo/MMM is DE-focused; the task doesn't state geography)	Medium — I planned as if this is the DE budget
Influencer contract floors allow a mix shift but not a fast exit	Medium — memo hints at floor-driven cuts (limitation #3)
H1 run-rates from the memo (§5.2) are accurate actuals	High
Brand's true ROI exceeds the measured 1.04× (8-week window)	Medium-high — standard MMM limitation, memo agrees

3.2 Unclear or unusual data points, and how I handled them *(code §5.3)*

1. **The memo optimises €16.1M while the task hands me €35M.** The single most important inconsistency — the "recommendation" I was given as primary input argues against my budget. Handled by treating €35M as a constraint, phasing toward Q4, and reserving budget for measurement (see §0).
2. **YouTube Paid Social: iROAS 37×, mROI "20 (capped)" on €0.07M spend (0.5% of PS).** A €2.82M contribution from €70K spend is not credible; the model itself caps it and flags LOW confidence. Handled: treated as "promising, unproven"; staged test allocation only.
3. **MMM vs attribution head-on conflict:** MMM ranks Influencer worst (mROI 0.40) while attribution shows Influencer as the volume-and-efficiency star (20K NCs at €56 CAC, CLV6M/CAC ≈ 100%); conversely TikTok looks great in MMM (1.35) and terrible in attribution (CAC €203, CLV/CAC 25%). Classic last-touch vs econometrics disagreement (influencers harvest credit; TikTok creates demand others harvest). Handled: this became Test #1 and #4 — I deliberately did not pick a side with money beyond what the constraint forces.
4. **"DE other (paid)": CAC €10.4 in May, 7K NCs from €73K.** Implausibly cheap and unexplained (branded search? retargeting? misattributed organic?). Excluded from decisions; flagged for definition.
5. **Attributed CAC is drifting up while volume drops** (DE total: 48K NCs @ €55.5 in March → 39K @ €54.6 in May, with April worse at €63): demand softening or attribution churn — matters for the H2 NC targets.
6. **CLV6M/CAC < 100% for META (46–56%) and TikTok (25–39%)** — on attribution numbers, paid social doesn't pay back in 6 months. Either CLV is longer-tailed than 6 months or paid social is structurally unprofitable on acquisition — a question the MMM answers differently (incremental revenue, not attributed NC). Handled: noted as a reason mROI, not CAC, drives my allocation.
7. **Brand iROAS 1.04× on an 8-week window** — a lower bound by construction. Handled: didn't cut brand below the fixed envelope, and biased the sub-mix toward the strongest measured line (Events 1.74×).
8. **Holdout R^2 = 0.29 / WAPE 27.7%** passes the stated gate but means point projections (±) are wide. Handled: used the model for *ranking and direction*, never for revenue promises; the memo's €57M revenue projection should carry explicit bands.
9. **Influencer L2 "floors bind"** (limitation #3): the per-platform cut is contract-driven, not optimal — so the L2 influencer numbers describe feasibility, not desirability. Handled: made contract renegotiation an explicit action item.

3.3 Missing business context I'd ask for

- **Why €35M?** The growth target, margin expectations, and whether "spend it all" beats "return €10M" — the plan changes completely depending on the answer.
- **Gross margin & unit economics per category** — MMM optimises revenue; the business optimises contribution. A revenue-optimal plan can be margin-destructive (especially promo-heavy Q4).
- **Geographic scope** — is the €35M DE-only (matching the MMM) or group-wide incl. FR/UK? The MMM covers only DE spend.
- **CLV beyond 6 months + cohort quality by channel** — to reconcile the CLV6M/CAC <1 problem and set true CAC ceilings.
- **Influencer contract structure** — which floors, until when, penalty costs; determines how fast the mix shift can happen.
- **Capacity/ops constraints** — Q1's syrup production gap showed fulfilment can throttle marketing; Black Week plans need ops sign-off (memo limitation #6 agrees).
- **Competitive & pricing calendar for H2** — a competitor's Black Week aggression changes the promo/media split.
- **New-product roadmap for H2** — launches (like syrup in Q1) create demand spikes the MMM attributes partly to media; knowing the calendar avoids over-crediting channels.
- **Attribution methodology** ("attributed NC" = last click? data-driven?) and the definition of "DE other (paid)".
- **Whether promo (€25M contribution, 18% of revenue) is inside or outside my €35M** — promo is modeled as a separate driver; if promo budget competes with media budget, the allocation question changes shape.

Time spent: ~1.5 hours (desk analysis of the memo + write-up). No code needed; all figures from the memo.

Appendix A — Data Quality Review

Systematic audit of `orders.csv`, `order_items.csv`, `products.csv` and `shipments.csv` (Apr 1 – May 31, 2026). Every issue below was verified with code; numbers are exact counts from the files as delivered. Issues are grouped by severity, and each one ends with the **question I would ask the interviewer / data owner**.

> **Reproducibility:** every check runs in **analysis.ipynb** — the `(code §…)` tag on each issue names the notebook section that produces the exact counts quoted.

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1. Schema mismatches vs. the case definition (ask first — cheap to clarify)

The files do not match the schemas described in the case brief:

Case brief says	Dataset actually has
`orders.csv` has `channel` ("Sales channel")	No `channel` column at all
`orders.csv` has `is_cancelled`	No `is_cancelled` column (cancellation only inferable from `shipments.shipment_status = 'cancelled'`)

Case brief says	Dataset actually has
`orders.csv` has `total_net_revenue`, `total_gross_revenue`	Columns are named `net_revenue`, `gross_revenue`, plus undocumented `customer_id`, `is_first_paid_order`, `shipping_revenue`, `refunded_value`
File `base_product.csv`	File is named `products.csv`
`shipments.csv` has only `order_id`, `delivered_at`	Has 9 columns: `shipment_status`, `delivery_status`, `warehouse`, `shipped_at`, `days_to_delivery`, `days_to_ship`, `is_on_time`, ...
Brief: "Syrup 3er Bundles (packs of **3 syrup bottles**...)"	`products.csv` names SKUs `11-00-43-0002-*` "Syrup **Pod** - 3er Bundle" — pods, not bottles

> **Ask:** Is the delivered dataset a newer export than the brief? Which document is authoritative? And is the 3er bundle (`11-00-43-0002%`) three *bottles* or three *pod packs*? This changes the interpretation of the retention/reorder analysis (a pod reorder = consumption refill; a bottle reorder = gifting/expansion).

How I handled it: I treated the CSVs as authoritative and used the brief only for business context (campaign dates, SKU lists, the May 12 reorder rule).

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2. High-impact issues (materially affect the analysis)

2.1 Duplicate orders — 500 duplicated `order_id`s in `orders.csv` *(code §1.1)*

1,000 rows carry 500 duplicated IDs. 445 pairs are *not* byte-identical — they differ slightly in `shipping_revenue` (e.g. 0.00 vs 0.03/0.05). Looks like an export dedup failure. **Handling:** kept the first occurrence. **Ask:** which extract is canonical, and is `shipping_revenue` reliable at all?

2.2 200 orphan order_items (`ORD-900001`–`ORD-900200`) *(code §1.2)*

200 line items reference order IDs that don't exist in `orders.csv` — and the ID range (900k) is far outside the normal range (~345k). Looks like injected test data or a truncated orders extract. **Handling:** excluded. **Ask:** are these test records, or is `orders.csv` missing rows?

2.3 No cancellation flag + re-shipment duplication in `shipments.csv` *(code §1.4)*

- `shipments.csv` has 366,218 rows for ~345k orders: **21,555 orders have 2–4 shipment rows** (typical pattern: shipped → cancelled → shipped again weeks later). Naively joining shipments to orders double-counts and mixes up delivery dates.
- 1,545 orders have *only* cancelled/ignored shipment rows (my proxy for a cancelled order); 270 orders have no shipment row at all.

Handling: used the *first delivered* shipment per order for delay analysis; flagged orders whose every shipment row is cancelled as "cancelled" and excluded them. **Ask:** what is the correct definition of a cancelled order? And do multiple shipment rows mean split shipments, replacements, or subscription-like repeat deliveries? (Many orders show two *successful* deliveries ~3–4 weeks apart, e.g. shipped Apr 23 and again May 18 — replacement shipments for the syrup production gap?)

2.4 Inconsistent net/gross revenue definitions in `orders.csv` *(code §1.1, §1.2)*

- **37,098 orders (10.8%) have `net_revenue` > `gross_revenue`.** In these rows `net_revenue ≈ gross_revenue + shipping_revenue` exactly — i.e. net *includes* shipping while gross excludes it, unlike all other rows.
- 764 orders have negative `net_revenue` with positive gross.
- Net/gross ratios are otherwise VAT-coherent (UK ≈ 1/1.20; DE mixes 7% food / 19% standard; FR mixes 5.5% / 20%), which confirms the general logic is right but the shipping treatment is inconsistent.
- Sum of `order_items.net_revenue` fails to reconcile with `orders.net_revenue` for **12.3% of orders** (gross reconciles almost perfectly: 149 mismatches).

Handling: I ran the whole analysis on **gross revenue**, which is internally consistent. **Ask:** what exactly is `net_revenue` net *of* (VAT? discounts? shipping in or out?), and why does it flip sign/definition on a subset of orders?

2.5 5,490 order-item rows with negative `net_revenue` but positive `gross_revenue` *(code §1.2)*

Including rows like gross €9.03 / net −€16.84. Possibly discount over-allocation on bundle decomposition. **Handling:** gross used throughout. **Ask:** is bundle discount allocation known to over-assign discounts to low-price components?

2.6 `shipments.is_on_time` looks broken *(code §1.4)*

344,406 rows are `False`, only 5,647 `True` — even 1–2-day deliveries are flagged not-on-time. Also `days_to_delivery` disagrees with `delivered_at - order_date` by >1 day in 145k rows (it appears to be measured from `shipped_at`, not order date). One row has `days_to_delivery = -1`. **Handling:** ignored `is_on_time` and `days_to_delivery`; computed delay as `delivered_at - order_date` myself. **Ask:** what SLA does `is_on_time` encode, and from which anchor timestamp is `days_to_delivery` counted?

2.7 13,724 order-item rows reference SKUs missing from `products.csv` *(code §1.2, §1.3)*

Top offenders:

SKU	Rows	Note
`10-00-42-0020`	7,069	**In the Syrup Bottle SKU range (10-00-42)**, sold only from launch day (Apr 19), avg €4.13 — looks like a syrup accessory (extra cap/pump?) that belongs in the campaign scope but is absent from both the brief and the product master
`11-00-01-0043-01-1`	2,225	Malformed suffix (`-1`) of a valid Energy SKU — €83.6k gross affected
`11-00-10-0139`	2,006	Component of milkshake bundles, missing from master
`11-00-03-0016-02` / `11-00-02-0006-02`	1,143 / 846	Missing from master

SKU	Rows	Note
`NOSYNC`	144	Placeholder; zero revenue; also flagged `is_bundle=True` with itself as component
`drako-1`, `raptor-1`, `syru-1`	17	Lowercase free-text SKUs that look like manually keyed syrup bottles ("drako" vs product name "Darko")

Handling: kept the rows (they carry real revenue) but they fall out of any category-level view. **Ask:** what is `10-00-42-0020`, and should the free-text/malformed SKUs be mapped back to real products?

2.8 Customer identity problems (breaks retention analysis at the margin) *(code §1.1)*

- 3,450 orders have **no `customer_id`** — these can never appear in reorder/retention metrics.
- **1,583 customers have more than one order flagged `is_first_order = TRUE`.**
- 10,137 orders are `is_first_order=TRUE` but `is_first_paid_order=FALSE` (plausible if the first order was free), yet 1,211 are the reverse — `is_first_order=FALSE` but `is_first_paid_order=TRUE` *and* that combination is hard to explain if flags are computed consistently.

Ask: how is customer identity resolved (email? account?), and are the first-order flags computed per shop or globally? Guest checkout = null customer_id?

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3. Anomalies that look like business events, not errors (need context) *(code §1.5)*

These change how the campaign windows should be read — I'd validate my reading with the team:

1. **May 18 spike:** 9,965 orders, but 55.7% have **zero gross revenue**, AOV €31, **72.3% first-time customers**, dominated by free Logo Shakers and Mixed Sachet Boxes. Looks like a **lead-gen / sampling giveaway**. It also coincides with a large wave of *re-shipments*. It inflates "orders" and depresses AOV and the new-customer quality picture for the post-period. → *Confirmed promo? Should zero-revenue orders be excluded from KPIs?*
2. **May 27 spike:** 12,159 orders, AOV €104 — driven by a **"Summer Cocktail 2026" launch** (ice cube trays, beach towels, cocktail-flavour tubs). A separate campaign that contaminates any "post-syrup-campaign baseline". → *Confirm so post-period comparisons can exclude it.*
3. **14,082 zero-gross orders overall** (only 87 of them cancelled) — freebies, replacements, or 100% discount codes?
4. **Pre-launch syrup sales:** 2 bottle orders and 2 pod orders dated Apr 16–18, before the Apr 19 launch (test orders? employees? early unlock?). Similarly, **155 zero-revenue 3er-bundle rows appear from Apr 19 onward**, before the official May 12 launch — they look like seeded samples/influencer sends. The brief's "count 3er as reorders" rule is unaffected (real sales clearly start May 12: 3,374 orders that day), but the early zero-value rows should be excluded from revenue.
5. **UK store is much smaller** (36k orders vs DE 177k / FR 132k) — sampled differently, or genuinely small market? Affects per-market conclusions.
6. **50,560 deliveries dated after May 31** (up to Jun 21) — expected, since late-May orders deliver in June; just note the shipment file has a longer horizon than the order file.

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4. Summary — top questions for the interviewer

1. What is `net_revenue` net of, and why does its definition flip (shipping included) for ~11% of orders? Which revenue field should steer the business?
2. How should a **cancelled order** be identified given there is no `is_cancelled` column?
3. Do multiple shipment rows per order represent **split shipments or replacements** (esp. the Apr→May re-ship wave around the syrup production gap)?
4. What are the **May 18** (zero-revenue sampling?) and **May 27** (Summer Cocktail launch) events, and should they be excluded from post-campaign baselines?
5. What is SKU **`10-00-42-0020`** (7k units, syrup range, missing from product master)?
6. Is the 3er bundle **pods or bottles**? (Brief and product master disagree.)
7. How is `customer_id` assigned (guest checkout → null?) and why do 1,583 customers have two "first" orders?
8. Which timestamp anchors `days_to_delivery`, and what SLA defines `is_on_time` (currently 98% False)?
9. Is the UK store's small size real or a sampling artifact?
10. Was there any **paid media / discount** behind the Apr 19 launch spike (34.5k orders, 6× normal)? Without spend data, campaign ROI can't be computed — only revenue uplift.

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5. Cleaning rules applied in my analysis *(code §2)*

Rule	Rows affected
Dedupe `orders` on `order_id` (keep first)	−500
Drop order_items with order IDs not in `orders`	−200
Exclude orders whose shipment rows are all cancelled/ignored	−1,545
Use **gross revenue** for all monetary metrics	—
Compute delivery delay as `delivered_at - order_date` (first delivered shipment)	—
Retention restricted to orders with a `customer_id`	53,738 of 55,487 bottle buyers identifiable

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Appendix B — Method & Project Blueprint

A map of how the case was approached, what was produced, and where to find everything. Read this first.

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1. Deliverables map

File	What it contains
Appendix B — Method	This document — the plan, method, and decision log

File	What it contains
Appendix A — Data Quality Review	Full audit of the 4 CSVs: verified errors, anomalies, cleaning rules applied, and the top 10 questions for the interviewer
Part 1	Question 1 — Syrup Campaign analysis, written as a briefing for the Head of Marketing (KPIs, uplift, adoption, delays, retention, cannibalisation, recommendations)
Part 2	Question 2 — €35M H2 budget allocation, incrementality testing roadmap, and strategic reflection on the MMM memo + attribution data
charts/	Four PNG charts referenced by the Q1 write-up
analysis.ipynb	The executable analysis: data audit (§1), cleaning rules (§2), every Q1 computation (§3), chart generation (§4), Q2 memo extraction & budget arithmetic (§5), further analyses beyond the brief (§6) — the .md files link into its sections

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2. Workflow (in execution order)

Phase 0 — Intake

1. Read `case-definition.md` (both questions, schemas, constraints, guiding questions).
2. Inventoried the project folder; extracted the CMO memo PDF to text (3 pages: MMM v5.3.1 results, H2 recommendation, DE attribution Mar–May 2026).
3. Previewed all four CSVs — **immediately found the schemas don't match the brief** (no `channel`/`is_cancelled` columns, extra columns, different file/field names), which promoted the data audit from a formality to a first-class workstream.

Phase 1 — Data quality audit (before any analysis)

Scripted checks in Python/pandas across all files: uniqueness, referential integrity, nulls, value ranges, cross-file reconciliation (order revenue vs. sum of items; shipments vs. orders), timestamp sanity, and campaign-specific validations (do the brief's SKUs exist? do 3er bundles really start May 12?). **Output:** Appendix A. The audit also produced the **cleaning rules** used downstream (dedupe, orphan removal, cancelled-order definition, "use gross revenue", "compute delivery delay manually").

Key catches that shaped the analysis:

- `net_revenue` is unreliable (definition flips for ~11% of orders; 5.5K negative item rows) → all analysis uses **gross**.
- `shipments` has re-shipment duplicates and a broken `is_on_time` → delay computed as first `delivered_at - order_date`.
- Two post-campaign business events (May 18 sampling giveaway, May 27 Summer Cocktail launch) contaminate the post-period → called out wherever post-period numbers are used.

Phase 2 — Question 1 analysis

Design decisions, then scripted analysis:

Decision	Choice	Why
Comparison windows	Pre Apr 1–18 · Campaign Apr 19–May 4 · Post May 5–31	Only pre-period available as baseline (first-ever product; dataset starts Apr 1)
Baseline metric	Per-day averages (windows have unequal lengths)	Comparability
Money metric	Gross revenue, ratios/relative deltas	Net unreliable; brief says don't trust absolutes
Syrup scope	Bottles `10-00-42-0001-*`, 10er pods `11-00-43-0001`, 3er `11-00-43-0002%` as reorders (per brief)	Brief definition; mystery SKU `10-00-42-0020` flagged but excluded pending clarification
Retention cohort	Campaign bottle buyers with a `customer_id`, first bottle order per customer, any later syrup purchase counts	Cleanest observable definition given identity gaps
Delay impact	Reorder rate by delivery-delay bucket of the first bottle order	Direct read of the production-gap question; time-truncation caveat stated
Cannibalisation	Energy €/day pre vs post + bottle-buyer cohort vs non-bottle existing-customer control	Cohort contrast controls for overall market growth

Analysis blocks executed: baseline & uplift (total, per market, excl. launch day) → new-vs-existing mix → market breakdown → daily trends → category mix → adoption (penetration, flavours, bundle structure, attach rates) → buying behaviour (AOV, units) → cross-purchase → delivery-delay impact → retention/reorders → cannibalisation → post-campaign event forensics (May 12/18/27 spikes).

Phase 3 — Charts

Four charts, built per the dataviz method (form chosen by the data's job, single-hue palette since every chart encodes one measure, direct labels, annotated events): daily revenue with campaign window shaded · daily bottle units (full + zoomed small-multiple, because the launch spike is 20× the baseline) · market adoption bars · **delay-vs-reorder bars (the money chart)**.

Phase 4 — Question 2 (desk analysis of the memo)

1. Extracted the memo PDF with `pypdf` and transcribed every table into DataFrames (L1/L2 iROAS & mROI, H1 run-rates, constraints, attribution) — notebook §5.1–5.2.
2. Identified the central tension: the task's **€35M** vs the MMM's own optimal **€16.1M** — made this the opening argument rather than hiding it.
3. Built the allocation from mROI ranking + saturation caveats + contract floors, with a testing reserve and Q3/Q4 phasing.
4. Designed the testing roadmap around the three biggest MMM-vs-attribution conflicts (Influencer, Google/Affiliate, TikTok/META), with sequencing and a Black-Week test freeze.

5. Wrote the strategic reflection: assumptions with confidence levels, 9 flagged data oddities and how each was handled, 10 missing-context asks.

Phase 5 — Further analyses (beyond the brief)

Seven additional analyses (notebook §6, write-up §5 of the Q1 briefing): refunds/cancellations, May 18 sampling conversion, new-customer cohort quality, warehouse split of the delays, habit-formation effect, a truncation-corrected delay-retention estimate (which **revised the headline lost-reorders figure from ~2,400 to ~1,000**), and a refill-demand planning number.

Phase 6 — Write-ups

Each deliverable written for its audience: Q1 as a marketing-leadership briefing (findings first, method in footnotes), Q2 as a planning document (logic per line item), data review as an engineering-style audit (issue → evidence → handling → question).

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3. Principles followed

- **Audit before analysis** — every headline number sits on the cleaning rules from Phase 1, and those rules are documented and reversible.
- **Data vs. interpretation kept separate** — write-ups label "what the data shows" vs. "my hypothesis" (per the case's explicit requirement).
- **Relative numbers over absolutes** — per the dataset disclaimer, all conclusions are ratios, shares and deltas.
- **Reproducibility** — everything numeric comes from analysis.ipynb (no spreadsheet hand-edits); the write-ups cite notebook sections (`code §…`) so any figure can be regenerated or challenged.
- **Say the uncomfortable thing** — the Q2 answer leads with the budget being ~2× model-optimal; the Q1 answer quantifies what the production gap cost in retention.

4. Known limitations of this work

- No media spend data for Q1 → uplift measured, ROI not measurable.
- Retention window is only 4–6 weeks post-purchase; 90-day retention and pods consumption cycles need June+ data.
- Post-campaign baseline is contaminated by two unrelated events (documented, not fully removable).
- Q2 assumes the MMM's response curves extrapolate to ~2× spend — explicitly flagged as low-confidence.
- Open data questions (see the data quality review) could shift details, though not the direction, of the findings.

5. Rough time log

Phase	Time
Intake + PDF/data exploration	~30 min
Data quality audit	~45 min
Q1 analysis + charts	~1.5 h

Phase	Time
Q2 desk analysis	~45 min
Further analyses	~1 h
Write-ups	~1.5 h
Total	**~6 h** (tool-assisted, fully scripted)